Aditya Wagh

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LinkedIn | GitHub | Portfolio

Professional Summary

AI engineer building agentic AI, RAG, and LLM systems, with a focus on making them reliable, tested, and measurable. Open-source contributor with a merged feature in openapi/gknowledge and founder of OpenWave India, an open-source community for practical AI tools. Former Research Intern at IIT Bombay (FOSSEE) and author of three papers published in IEEE Xplore.

Work Experience

Co-Founder *May 2026–Present*
Nailed It Bro, India (Remote)

- Co-founded a studio that builds AI automations, web applications, websites, and QR-based systems for small businesses.
- Build full-stack applications with React, TypeScript, and Python, integrating LLM APIs into client workflows.

Research Intern *Dec 2025–Feb 2026*
Indian Institute of Technology (IIT) Bombay – FOSSEE

- Contributed Python and PySide6 modules to Osdag Bridge, the bridge analysis and design module of IIT Bombay's open-source structural engineering software.
- Built desktop UI workflows for bridge design to improve usability for engineering users, collaborating through code reviews with domain experts and senior developers.

Project Developer Intern *Jan 2025–Jun 2025*
BigBuddy Foundation, India

- Developed an OCR-to-Excel automation system using Tesseract OCR and OpenAI LLMs to extract information from unstructured documents, reducing manual processing effort by approximately 70%.
- Built and deployed a MERN-stack educational platform supporting 100+ users with interactive learning features.

Open Source

openapi/gknowledge – Merged Pull Request #46 *Oct 2026*
GitHub

- Implemented local Ollama model discovery in the Preferences UI of a GNOME AI knowledge app, using background-thread detection with a 2-second timeout, GTK updates via `GLib.idle_add`, and no document content sent during discovery.
- Added mocked tests for successful discovery, connection failures, timeouts, and malformed responses; all CI checks (tests, Debian, Flatpak) passed.

OpenWave India – Founder *Oct 2026–Present*
GitHub

- Started an open-source GitHub community where developers can pick up issues and collaborate on practical AI tools.

Selected Projects

Veriflow – Verification Skill for AI Coding Agents *2026*
GitHub

- Built an open-source Claude Code skill that discovers a project's real test, lint, type-check, and build commands across Python, JavaScript/TypeScript, Rust, Go, Java, and .NET, and runs them only on opt-in with per-command timeouts.
- Reports structured outcomes (PASS, FAIL, BLOCKED, UNKNOWN) with evidence, caps repair attempts at three, and adds static security rules and dependency checks for hallucinated packages.

DocExtract Pro – AI Invoice Extraction for Indian GST *2026*
GitHub

- Built an invoice pipeline using OpenAI Structured Outputs with a rule-based fallback, PaddleOCR for scanned pages, and GSTIN checksum and tax-arithmetic validation.
- Added human review, Excel/CSV/Tally XML export, and a WhatsApp intake webhook; deployed with Docker using Flask, React, and SQLite.

RAGBench – RAG Evaluation & Optimization Framework *2026 (in progress)*
GitHub

- Designing an open-source framework to benchmark RAG pipelines by comparing chunking, embedding models, rerankers, and retrieval strategies on fixed datasets, with retrieval precision/recall and faithfulness metrics.

NeuroCode AI – Code Intelligence Platform *2026 (in progress)*
GitHub

- Building a multi-language code analysis pipeline using Tree-sitter and AST parsing to extract code structure from repositories, as the foundation for semantic code search and repository-aware question answering.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, Java, C++
AI / LLM: LLM APIs (OpenAI), Retrieval-Augmented Generation (RAG), Embeddings & Vector Search, Agentic AI & Tool Calling, Structured Outputs, Prompt Engineering, Hugging Face Transformers, Scikit-learn, NLP
Backend & APIs: FastAPI, Flask, REST APIs, Node.js, Express.js
Frontend: React, Tailwind CSS
Data & Infrastructure: PostgreSQL (Supabase), MongoDB, MySQL, SQLite, Docker, Git, Linux
Testing & Quality: pytest, mocking, CI (GitHub Actions)

Languages

English (professional working proficiency) | German (A2, working towards B1)

Education

B.Tech. in Computer Science and Engineering (AI & ML) *2022–2026*
Vishwakarma Institute of Information Technology (VIIT), Pune, India **CGPA: 8.42/10**

Publications

- **Contextual Filler Word Detection using BERT and Prosodic Features**, ICCUBEA 2025 – Published in IEEE Xplore.
- **AI Agent for Scientific Paper Analysis**, INSPECT 2025 – Published in IEEE Xplore.
- **Robust Facial Expression Recognition with VGG16**, APCIT 2025 – Published in IEEE Xplore.
- **Bridging the Language Gap: A Survey and Comparative Analysis of Open-Source Chatbots for Indian Languages**, ICRRAI 2025 – Accepted for publication.

Certifications

- Building RAG Applications Using MongoDB – MongoDB, 2025.
- IBM AI Engineering Specialization – Coursera, 2024.
- IBM Full Stack Software Developer – Coursera, 2024.

Additional Information

- Completed the CDAC UAV Bootcamp on AI and Drone Systems (2025).
- Global Immersion Program, UCSI University Malaysia (2025), covering AI-driven cybersecurity, data privacy, and ethical AI.